

The Inverse-Square Law of Gravity

PhET Gravity Force Lab · PhET

Senior Secondary (Years 11-12)	~30 minutes	Physics
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NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site’s curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/gravity-force-lab>
Curriculum links: forces · gravity

Learning intentions

- I can describe how gravitational force changes as mass changes.
- I can describe how gravitational force changes as distance changes.
- I can use collected data to verify the inverse-square relationship between force and distance.

Getting started

Open Gravity Force Lab. Start with the two default masses at a fixed distance apart and note the force reading. You'll then vary mass and distance independently in the tasks below, so change only one variable at a time and keep a record of your readings.

Tasks

1. Record the initial force reading with the default masses and separation.

2. Keeping distance fixed, double one mass twice in a row (e.g. 1x, 2x, 4x the original) and record the force each time.

Mass multiplier	Force reading

3. Based on your table, describe in words how force depends on mass.

4. Return mass to its original value. Now keep mass fixed and increase the distance between the masses (e.g. 1x, 2x, 4x the original separation), recording the force each time.

Distance multiplier	Force reading

5. Using your distance data, calculate the ratio of forces when distance doubles, and again when distance quadruples. Compare this to what you'd expect if force were proportional to $1/\text{distance}^2$ versus $1/\text{distance}$.

Working:

6. Explain what 'inverse-square' means using your own results, and why doubling the distance doesn't simply halve the force.

7. Use the inverse-square relationship to explain why astronauts in low Earth orbit still experience most of Earth's gravity, despite the popular idea that there is 'no gravity in space'.

Extension (optional)

Two planets have the same mass, but one has twice the radius of the other. Using what you've learned about the inverse-square law, explain which planet would have the smaller surface gravity, and why.
